

A-Level Maths:

1st class



Integration 2

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- Integration by substitution
- Integration by parts

With practice, the technique of **integration by recognition** can save a lot of time.

However, when it is too difficult to use integration by recognition we can use a more formal method of reversing the chain rule called **integration by substitution**.

To see how this method works consider the integral $\int (5x + 2)^3 dx$.

Let $u = 5x + 2$ so that $\int (5x + 2)^3 dx = \int u^3 dx$

The problem now is that we can't integrate a function in u with respect to x . We therefore need to write dx in terms of du .

$$u = 5x + 2 \Rightarrow \frac{du}{dx} = 5$$

When we used the chain rule for differentiation we saw that we can treat $\frac{du}{dx}$ informally as a fraction, so:

$$\frac{du}{dx} = 5 \Rightarrow du = 5dx \Rightarrow dx = \frac{1}{5} du$$

So if $u = 5x + 2$ and $dx = \frac{1}{5} du$

$$\begin{aligned} \int (5x + 2)^3 dx &= \int u^3 \frac{1}{5} du \\ &= \frac{1}{20} u^4 + c \end{aligned}$$

$$\int (5x + 2)^3 dx = \frac{1}{20} (5x + 2)^4 + c$$

Reversing the chain rule for logarithmic functions

Use a suitable substitution to find $\int x(2x^2 - 5)^5 dx$.

$$\begin{aligned}\text{Let } u &= 2x^2 - 5 & \frac{du}{dx} &= 4x \\ & & \frac{dx}{du} &= \frac{1}{4x} \\ & & dx &= \frac{1}{4x} du\end{aligned}$$

Substituting u and dx into the original problem gives:

$$\begin{aligned}\int x(2x^2 - 5)^5 dx &= \int x u^5 \frac{1}{4x} du \\ &= \frac{1}{4} \int u^5 du \\ &= \frac{1}{24} u^6 + c\end{aligned}$$

Notice that the x 's cancel out.

$$\int x(2x^2 - 5)^5 dx = \frac{1}{24}(2x^2 - 5)^6 + c$$

This integral could also have been found directly by recognition.

However, there are functions that can be integrated by use of a suitable substitution but not by recognition. For instance:

Use the substitution $u = 1 - 2x$ to find $\int x(1 - 2x)^4 dx$.

If $u = 1 - 2x$ then $\frac{du}{dx} = -2$

$$\frac{dx}{du} = -\frac{1}{2}$$

$$dx = -\frac{1}{2} du$$

Also if $u = 1 - 2x$ then

$$x = \frac{1}{2}(1 - u)$$

We also have to substitute the x so that the whole integrand is in terms of u .

Substituting these into the original problem gives:

$$\begin{aligned}\int x(1 - 2x)^4 dx &= \int \frac{1}{2}(1 - u)u^4 \left(-\frac{1}{2}\right) du \\ &= -\frac{1}{4} \int (1 - u)u^4 du \\ &= -\frac{1}{4} \int (u^4 - u^5) du\end{aligned}$$



$$\begin{aligned}-\frac{1}{4} \int (u^4 - u^5) du &= -\frac{1}{4} \left(\frac{u^5}{5} - \frac{u^6}{6} \right) + c \\&= -\frac{1}{4} \left(\frac{6u^5 - 5u^6}{30} \right) + c \\&= -\frac{1}{120} u^5 (6 - 5u) + c\end{aligned}$$

$$\begin{aligned}\int x(1-2x)^4 dx &= -\frac{1}{120} (1-2x)^5 (6 - 5(1-2x)) + c \\&= -\frac{1}{120} (1-2x)^5 (10x+1) + c \\&= \frac{1}{120} (2x-1)^5 (10x+1) + c\end{aligned}$$

Definite integration by substitution

Use the substitution $u = \sqrt{8 - x}$ to find the area under the curve $y = \frac{2x}{\sqrt{8 - x}}$ between $x = 4$ and $x = 7$.

If $u = (8 - x)^{\frac{1}{2}}$ then $\frac{du}{dx} = \frac{1}{2}(8 - x)^{-\frac{1}{2}}(-1)$

Using the chain rule for differentiation.

$$= -\frac{1}{2(8 - x)^{\frac{1}{2}}}$$

$$= -\frac{1}{2u}$$

So $\frac{dx}{du} = -2u \Rightarrow dx = -2u du$

Definite integration by substitution

Now we need to find x in terms of u .

If $u = (8 - x)^{\frac{1}{2}}$ then $u^2 = 8 - x$

$$x = 8 - u^2$$

Rewrite the limits in terms of u :

when $x = 3$, $u = \sqrt{8 - 3} = 2$

when $x = 1$, $u = \sqrt{8 - 1} = 1$

The area is given by $\int_4^7 \frac{2x}{\sqrt{8-x}} dx$. Rewrite this in terms of u :

$$\begin{aligned} \int_4^7 \frac{2x}{\sqrt{8-x}} dx &= \int_1^2 \frac{2(8-u^2)}{u} (-2u) du \\ &= \int_1^2 -4(8-u^2) du \\ &= 4 \int_1^2 (u^2 - 8) du \end{aligned}$$

$$\begin{aligned}4 \int_1^2 (u^2 - 8) du &= 4 \left[\frac{u^3}{3} - 8u \right]_1^2 \\&= 4 \left(\frac{8}{3} - 16 - \frac{1}{3} + 8 \right) \\&= 4 \left(\frac{7}{3} - 8 \right) \\&= -22\frac{2}{3}\end{aligned}$$

Therefore, the required area is $22\frac{2}{3}$ units squared.



- Integration by parts



Suppose we wish to integrate the product of two functions, such as $x \sin x$, where one of the functions is not related to the derivative of the other.

An expression such as this can be integrated using the method of **integration by parts**.

When we *differentiate* the product of two functions u and v we use the product rule:

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

where u and v are functions of x .

Integrating throughout with respect to x gives:

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

This can be rearranged to give:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

To integrate a product using this formula we let one part equal u and the other equal $\frac{dv}{dx}$.

We find $\frac{du}{dx}$ by differentiating the part we called u .

We find v by integrating the part we called $\frac{dv}{dx}$.

It is important to choose u and $\frac{dv}{dx}$ so that

$\int v \frac{du}{dx} dx$ is easier to integrate than $\int u \frac{dv}{dx} dx$.



Integration by parts

So, to integrate $x \sin x$ with respect to x :

differentiate $\left(\begin{array}{l} u = x \\ \frac{du}{dx} = 1 \end{array} \right.$ and $\left(\begin{array}{l} \frac{dv}{dx} = \sin x \\ v = -\cos x \end{array} \right.$ integrate

Now, using the formula $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

We don't need the "+ c" here.

$$\begin{aligned} \int x \sin x dx &= -x \cos x - \int -\cos x dx \\ &= -x \cos x - -\sin x + c \\ &= \sin x - x \cos x + c \end{aligned}$$

Find $\int xe^{2x} dx$.

$$u = x \quad \text{and} \quad \frac{dv}{dx} = e^{2x}$$

$$\frac{du}{dx} = 1 \quad \text{and} \quad v = \frac{e^{2x}}{2}$$

Now, apply the formula $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

$$\begin{aligned} \int xe^{2x} dx &= x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \\ &= \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + c \\ &= \frac{e^{2x}}{4} (2x - 1) + c \end{aligned}$$

Integration by parts

Find $\int 8x \ln x \, dx$.

We don't know the integral of $\ln x$ so:

$$\begin{array}{lll} u = \ln x & \text{and} & \frac{dv}{dx} = 8x \\ \frac{du}{dx} = \frac{1}{x} & \text{and} & v = 4x^2 \end{array}$$

Now, using the formula, $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$:

$$\begin{aligned} \int 8x \ln x \, dx &= 4x^2 \ln x - \int 4x^2 \times \frac{1}{x} dx \\ &= 4x^2 \ln x - \int 4x \, dx \\ &= 4x^2 \ln x - 2x^2 + c \\ &= 2x^2(2\ln x - 1) + c \end{aligned}$$

The integral of $\ln x$

We can also use integration by parts to find the integral of $\ln x$.

We do this by writing $\ln x$ as $(1 \times \ln x)$.

$$\begin{array}{ll} u = \ln x & \text{and} \quad \frac{dv}{dx} = 1 \\ \frac{du}{dx} = \frac{1}{x} & \text{and} \quad v = x \end{array}$$

Now we can integrate by parts:

$$\begin{aligned} \int 1 \times \ln x \, dx &= x \ln x - \int x \frac{1}{x} \, dx \\ &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + c \end{aligned}$$

$$\int \ln x \, dx = x \ln x - x + c$$

Integration by parts

To evaluate a definite integral using integration by parts we use:

$$\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b v \frac{du}{dx} dx$$

Evaluate $\int_0^{\frac{\pi}{2}} 2x \cos x dx$

$$u = 2x \quad \text{and} \quad \frac{dv}{dx} = \cos x$$

$$\frac{du}{dx} = 2 \quad \text{and} \quad v = \sin x$$

$$\int_0^{\frac{\pi}{2}} 2x \cos x dx = [2x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2 \sin x dx$$

$$\sin \frac{\pi}{2} = 1$$

$$\begin{aligned} \cos \frac{\pi}{2} &= 0 \\ \cos 0 &= 1 \end{aligned}$$

$$= (\pi \sin \frac{\pi}{2} - 0) - [-2 \cos x]_0^{\frac{\pi}{2}}$$

$$= \pi - (-2 \cos \frac{\pi}{2} + 2 \cos 0)$$

$$= \pi - 2$$